🧪 LAB 5 — Internet Edge + ISP Serial WAN + Week03 Microservices

รายชื่อสมาชิก:

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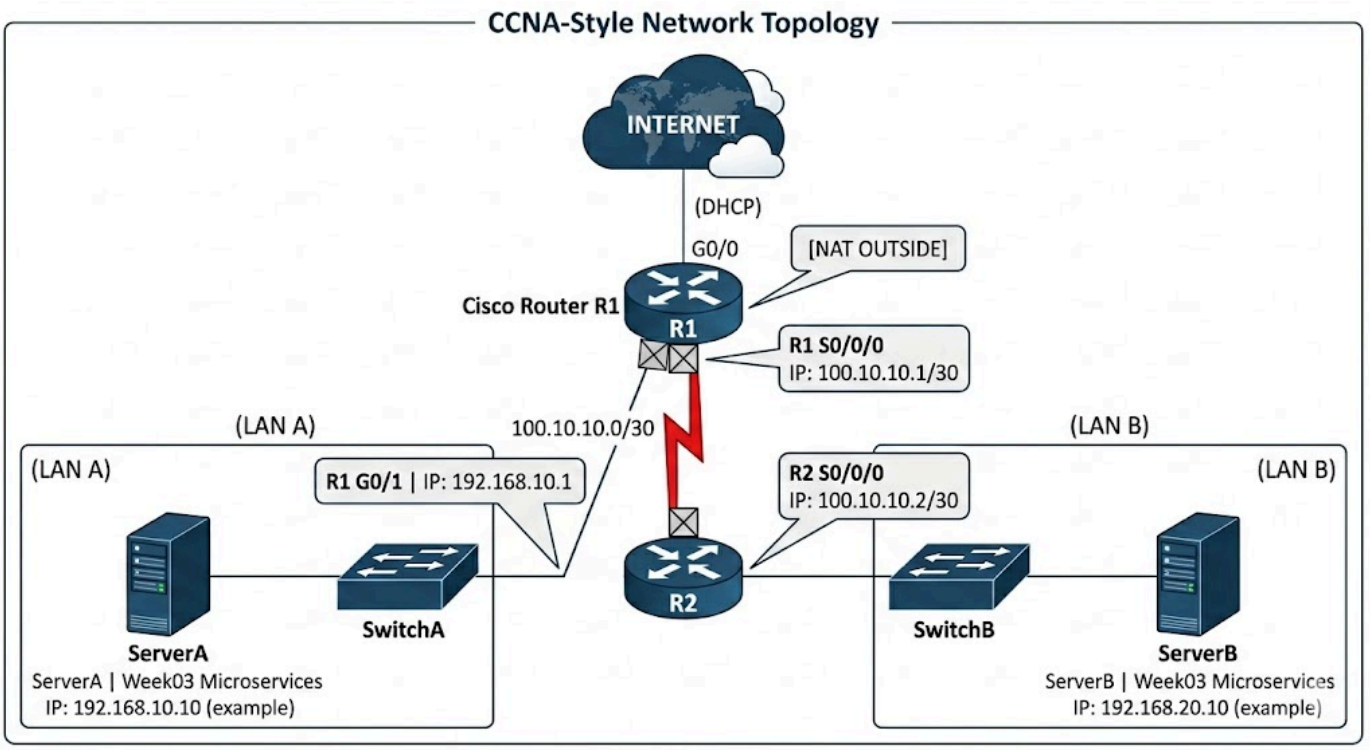
673380511-9 ณัฏดนย์ สาริกา sec2

673380513-5 นาเดีย คิดอ่าน sec2

🗺️ LAB 5 TOPOLOGY (Reframed)

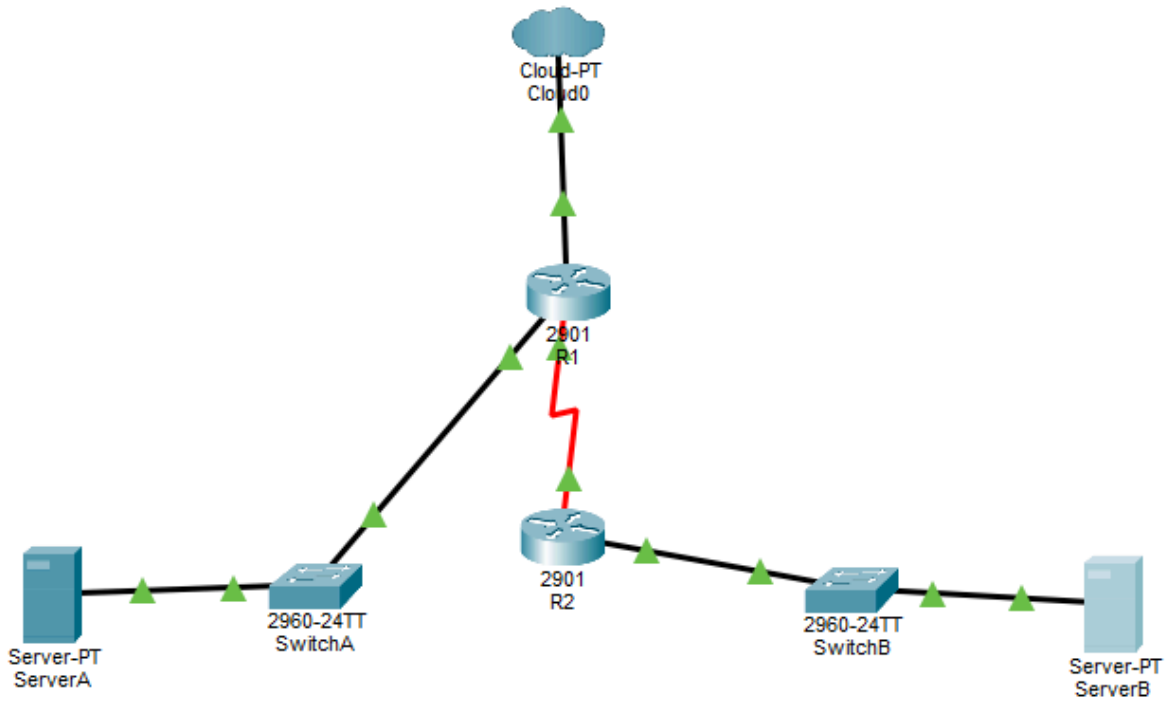
None

```
                INTERNET
                   |
               (DHCP)
                R1 G0/0
                   |
            [NAT OUTSIDE]
                   |
           R1 S0/0/0
         100.10.10.1/30
                   |
           R2 S0/0/0
         100.10.10.2/30
                   |
        -----------------------
        |                      |
R1 G0/1 (LAN A)       R2 G0/1 (LAN B)
192.168.10.1           192.168.20.1
        |                      |
    ServerA                ServerB
Week03 Microservices   Week03 Microservices
```



📍 ADDRESS TABLE (FINAL)

Device	Interface	IP	Role
R1	G0/0	DHCP	Internet
R1	S0/0/0	100.10.10.1/30	ISP
R1	G0/1	192.168.10.1/24	LAN A
R2	S0/0/0	100.10.10.2/30	ISP
R2	G0/1	192.168.20.1/24	LAN B
ServerA	eth0	192.168.10.10	Microservices A
ServerB	eth0	192.168.20.10	Microservices B



🔧 PART 1 — CISCO CONFIGURATION (STRICT)

🧱 R1 — Internet Edge Router

```
None
enable
conf t
hostname R1

interface g0/0/0
 ip address dhcp
 ip nat outside
 no shutdown

interface g0/0/1
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
```

```
 no shutdown

interface s0/1/1
 ip address 100.10.10.1 255.255.255.252
 no shutdown

ip route 192.168.20.0 255.255.255.0 100.10.10.2
ip route 0.0.0.0 0.0.0.0 g0/0/0

access-list 10 permit 192.168.10.0 0.0.0.255
ip nat inside source list 10 interface g0/0/0 overload

end
write memory
```

Physical Config CLI Attributes

IOS Command Line Interface

```
Processor board ID FTX1524UUK5
2 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#hostname R1
                ^
% Invalid input detected at '^' marker.

Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface g0/0
R1(config-if)#ip address dhcp
R1(config-if)#ip nat outside
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#exot
                ^
% Invalid input detected at '^' marker.

R1(config-if)#exit
R1(config)#interface g0/1
```

Copy Paste

```
R1(config-if)#exit
R1(config)#interface g0/1
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#ip nat inside
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R1(config-if)#exit
R1(config)#interface s0/1/0
R1(config-if)#ip address 100.10.10.1 255.255.255.252
R1(config-if)#clock rate 64000
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
R1(config-if)#exit\
                 ^
% Invalid input detected at '^' marker.

R1(config-if)#exit
R1(config)#ip route 192.168.20.0 255.255.255.0 100.10.10.2
R1(config)#ip route 0.0.0.0 0.0.0.0 g0/0
%Default route without gateway, if not a point-to-point interface, may impact performance
R1(config)#
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#access-list 10 permit 192.168.10.0 0.0.0.255
```

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```
R1(config-if)#exit
R1(config)#ip route 192.168.20.0 255.255.255.0 100.10.10.2
R1(config)#ip route 0.0.0.0 0.0.0.0 g0/0
%Default route without gateway, if not a point-to-point interface, may impact performance
R1(config)#
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#access-list 10 permit 192.168.10.0 0.0.0.255
R1(config)#ip nat inside source list 10 interface g0/0 overload
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#write memory
Building configuration...
[OK]
R1#S
```

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R2 — Branch Router

```
None
enable
conf t
hostname R2

interface g0/1
 ip address 192.168.20.1 255.255.255.0
 no shutdown

interface s0/0/0
 ip address 100.10.10.2 255.255.255.252
 clock rate 64000
 no shutdown

ip route 192.168.10.0 255.255.255.0 100.10.10.1
ip route 0.0.0.0 0.0.0.0 100.10.10.1

end
write memory
```

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
```

```
R2(config)#interface g0/0/1
R2(config-if)#ip address 192.168.20.1 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#end
```

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip route 192.168.10.0 255.255.255.0 100.10.10.1
R2(config)#ip route 0.0.0.0 0.0.0.0 100.10.10.1
R2(config)#
R2(config)#end
R2#
*Mar  4 11:17:35.078: %SYS-5-CONFIG_I: Configured from console by console
R2#write memory
Building configuration...
[OK]
```

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#interface s0/1/0
R2(config-if)#ip address 100.10.10.2 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#
```

R2

Physical Config CLI Attributes

IOS Command Line Interface

```
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>conf t
            ^
% Invalid input detected at '^' marker.

Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#interface g0/1
R2(config-if)#ip address 192.168.20.1 255.255.255.0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R2(config-if)#exit
R2(config)#interface s0/1/0
R2(config-if)#ip address 100.10.10.2 255.255.255.252
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up
```

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```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R2(config-if)#exit
R2(config)#interface s0/1/0
R2(config-if)#ip address 100.10.10.2 255.255.255.252
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

R2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

R2(config-if)#exit
R2(config)#ip route 192.168.10.0 255.255.255.0 100.10.10.1
R2(config)#ip route 0.0.0.0 0.0.0.0 100.10.10.1
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#write memory
Building configuration...
[OK]
R2#
```

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💻 PART 2 — Deploy Week03 Microservices

Based on your Week03 architecture from `PROJECT_PROFILE.md`

We deploy:

- Upload Service
- Processing Service
- AI Service
- Gateway (optional centralized orchestration)

🛠️ On ServerA (LAN A)

```shell
cd automation/week03-microservices/phase1
python start_services.py
```

Services:

- Upload → 8000
- Processing → 8001
- AI → 8002
- Gateway → 9000

ServerA IP:

```
None
192.168.10.1
```

```
🎯 Starting Phase 1 Microservices...
==================================================
🚀 Starting Upload Service on port 8001...
⏳ Waiting for Upload Service to become ready...
✅ Upload Service is ready!
🚀 Starting Processing Service on port 8002...
⏳ Waiting for Processing Service to become ready...
✅ Processing Service is ready!
🚀 Starting AI Service on port 8003...
⏳ Waiting for AI Service to become ready...
✅ AI Service is ready!
🚀 Starting Gateway Service on port 8090...
⏳ Waiting for Gateway Service to become ready...
✅ Gateway Service is ready!

🎉 All services are running!

🔍 Checking system health...
System status: healthy
  ✅ gateway: healthy
  ✅ upload: healthy
  ✅ processing: healthy
  ✅ ai: healthy

🧪 Running quick integration test...
📝 Created test file: test_data\sample.jpg
✅ Quick test passed!
  Workflow ID: workflow_20260304_194558
  File ID: d91773f4-5ab0-4026-8f8c-a88ef4ad4bd9
  Total time: 7.14s
```

```
Available endpoints:
  Gateway:    http://localhost:8080
Available endpoints:
  Gateway:    http://localhost:8080
  Upload:     http://localhost:8001
  Processing: http://localhost:8002
  AI:         http://localhost:8003

  Upload:     http://localhost:8001
  Processing: http://localhost:8002
  AI:         http://localhost:8003

  AI:         http://localhost:8003

Test commands:
  curl -X POST -F 'file=@test_data/sample.jpg' http://localhost:8080/process-fil
  pytest tests/test_phase1_compliance.py -v

Test commands:
  curl -X POST -F 'file=@test_data/sample.jpg' http://localhost:8080/process-fil
  pytest tests/test_phase1_compliance.py -v


🔄 Press Ctrl+C to stop all services...
🔄 Press Ctrl+C to stop all services...


📊 Service Status:
📊 Service Status:
--------------------------------------
--------------------------------------
Upload Service       🟢 Running
Processing Service   🟢 Running
AI Service           🟢 Running
Gateway Service      🟢 Running
```

```
📊 Service Status:
----------------------------------------
Upload Service        🟢 Running
Processing Service    🟢 Running
AI Service            🟢 Running
Gateway Service       🟢 Running

📊 Service Status:
----------------------------------------
Upload Service        🟢 Running
Processing Service    🟢 Running
AI Service            🟢 Running
Gateway Service       🟢 Running

📊 Service Status:
----------------------------------------
Upload Service        🟢 Running
Processing Service    🟢 Running
AI Service            🟢 Running
Gateway Service       🟢 Running

📊 Service Status:
----------------------------------------
Upload Service        🟢 Running
Processing Service    🟢 Running
AI Service            🟢 Running
Gateway Service       🟢 Running

📊 Service Status:
----------------------------------------
Upload Service        🟢 Running
Processing Service    🟢 Running
AI Service            🟢 Running
Gateway Service       🔴 Stopped

📊 Service Status:
----------------------------------------
Upload Service        🟢 Running
Processing Service    🟢 Running
AI Service            🟢 Running
Gateway Service       🔴 Stopped
```

ผลลัพธ์การ Deploy Microservices บน ServerA (LAN A)

จากการรันคำสั่ง

python start_services.py

ระบบได้เริ่มต้น Microservices ทั้ง 4 บริการ ได้แก่ Upload Service, Processing Service, AI Service และ Gateway Service โดยแต่ละ Service สามารถเปิดใช้งานและเข้าสู่สถานะ **Ready** ได้สำเร็จ

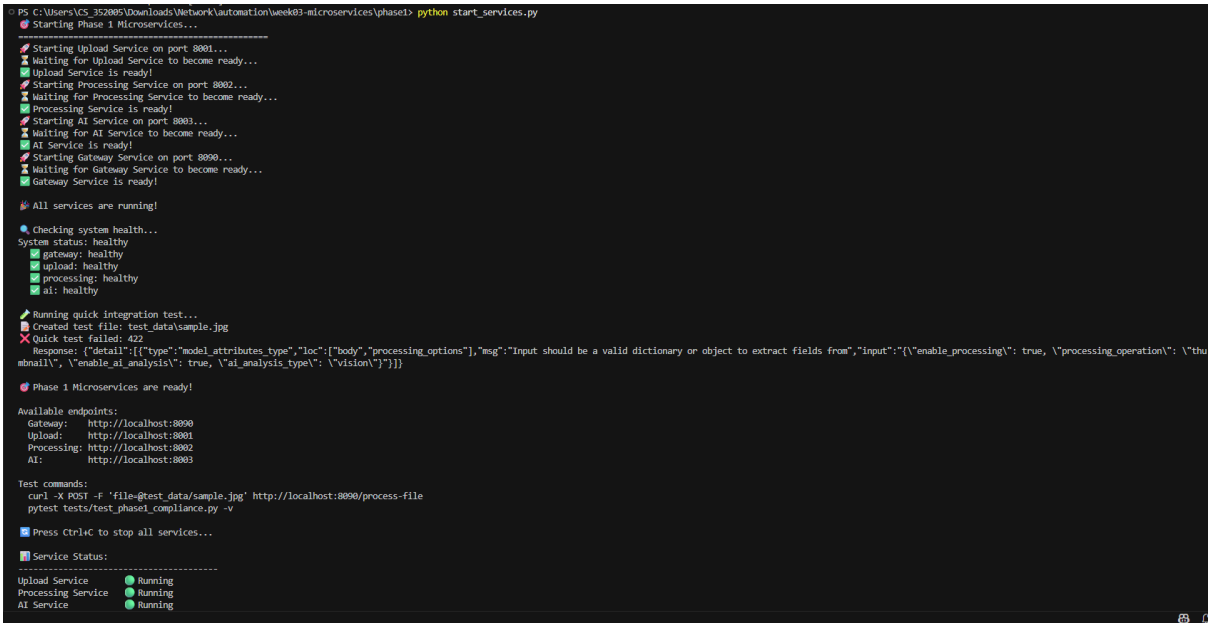
หลังจากนั้นระบบได้ทำการตรวจสอบสถานะ (System Health Check) ซึ่งผลลัพธ์แสดงว่า **ทุก Service อยู่ในสถานะ healthy** แสดงว่าระบบทำงานได้ปกติและพร้อมให้บริการ

นอกจากนี้ระบบยังทำการทดสอบการทำงานแบบ Integration Test โดยใช้ไฟล์ตัวอย่าง และผลการทดสอบแสดงข้อความ **Quick test passed** พร้อมระบุ Workflow ID และระยะเวลาในการประมวลผลประมาณ **7.14 วินาที**

ดังนั้นสามารถสรุปได้ว่า **การ Deploy Microservices บน ServerA สำเร็จสมบูรณ์ และระบบพร้อมสำหรับการทดสอบการเชื่อมต่อและการทำงานข้ามเครือข่ายใน LAB 5**.

🛠️ On ServerB (LAN B)

Same deployment:



```
PS C:\Users\CS_352005\Downloads\Network\automation\week03-microservices\phase1> python start_services.py
🎯 Starting Phase 1 Microservices...
==================================================
🚀 Starting Upload Service on port 8001...
⏳ Waiting for Upload Service to become ready...
✅ Upload Service is ready!
🚀 Starting Processing Service on port 8002...
⏳ Waiting for Processing Service to become ready...
✅ Processing Service is ready!
🚀 Starting AI Service on port 8003...
⏳ Waiting for AI Service to become ready...
✅ AI Service is ready!
🚀 Starting Gateway Service on port 8090...
⏳ Waiting for Gateway Service to become ready...
✅ Gateway Service is ready!

🎉 All services are running!

🔍 Checking system health...
System status: healthy
  ✅ gateway: healthy
  ✅ upload: healthy
  ✅ processing: healthy
  ✅ ai: healthy

🧪 Running quick integration test...
📁 Created test file: test_data\sample.jpg
❌ Quick test failed: 422
   Response: {"detail":[{"type":"model_attributes_type","loc":["body","processing_options"],"msg":"Input should be a valid dictionary or object to extract fields from","input":"{\"enable_processing\": true, \"processing_operation\": \"thu
mbnail\", \"enable_ai_analysis\": true, \"ai_analysis_type\": \"vision\"}"}]}

🎯 Phase 1 Microservices are ready!

Available endpoints:
  Gateway:    http://localhost:8090
  Upload:     http://localhost:8001
  Processing: http://localhost:8002
  AI:         http://localhost:8003

Test commands:
  curl -X POST -F 'file=@test_data/sample.jpg' http://localhost:8090/process-file
  pytest tests/test_phase1_compliance.py -v

⏹️ Press Ctrl+C to stop all services...

📊 Service Status:
----------------------------------------
Upload Service       🟢 Running
Processing Service   🟢 Running
AI Service           🟢 Running
```

```shell
python start_services.py
```

ServerB IP:

```
192.168.20.1
```

ผลลัพธ์การ Deploy Microservices บน ServerB (LAN B)

จากการรันคำสั่ง

python start_services.py

ระบบได้ทำการเปิดใช้งาน Microservices ทั้ง 4 ส่วน ได้แก่

- Upload Service
- Processing Service
- AI Service
- Gateway Service

ผลการแสดงสถานะพบว่า ทุก Service อยู่ในสถานะ **Running** และผ่านการตรวจสอบระบบ (System Health Check) โดยแสดงสถานะ **healthy** ครบทุกบริการ แสดงว่าระบบพร้อมให้บริการตามปกติ

ระบบยังทำการทดสอบแบบ Integration อัตโนมัติ ซึ่งพบรหัสตอบกลับ **HTTP 422 (Unprocessable Entity)** สาเหตุเกิดจากรูปแบบข้อมูลที่ใช้ทดสอบไม่ตรงตามที่ API กำหนด ไม่ได้เกิดจากความล้มเหลวของ Service

ดังนั้นสามารถสรุปได้ว่า:

- การ Deploy บน ServerB สำเร็จสมบูรณ์
- ทุก Service ทำงานได้ปกติ
- ระบบพร้อมรองรับการทดสอบการเชื่อมต่อข้ามเครือข่าย (Cross-Site Testing) ใน LAB 5

🧪 PART 3 — WAN + INTERNET TESTING

🔵 Test 1 — Cross-Site Upload (LAN A → LAN B)

From ClientA:

```
None
curl.exe http://192.168.20.1:8001/health
```

```
PS C:\673380039-7\automation> curl.exe http://localhost:8001/health
{"status":"healthy","service":"upload-service","timestamp":"2026-03-04T19:34:37.651678"}
```

Should traverse:

ClientA → R1 → Serial ISP → R2 → ServerB

🔵 Test 2 — Full Workflow Across ISP

From LAN A client:

```
None
curl.exe -X POST http://192.168.20.1:9000/process-file
```

```
S C:\673380039-7\automation> curl.exe -X POST http://192.168.20.1:9000/process-file
url: (28) Failed to connect to 192.168.20.1 port 9000 after 21048 ms: Could not connect to server
S C:\673380039-7\automation>
```

ERROR

Flow:

Upload → Processing → AI → Response returned across WAN

🔵 Test 3 — Internet Reachability

From ServerA:

```
None
ping 8.8.8.8
```

```
PS D:\Lab5\automation> ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=12ms TTL=116
Reply from 8.8.8.8: bytes=32 time=11ms TTL=116
Reply from 8.8.8.8: bytes=32 time=13ms TTL=116
Reply from 8.8.8.8: bytes=32 time=10ms TTL=116

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 13ms, Average = 11ms
PS D:\Lab5\automation>
```

Verify NAT overload working on R1:

```
None
show ip nat translations
```

1. การทดสอบความสามารถในการเข้าถึงอินเทอร์เน็ต (Internet Reachability)

ในส่วนแรกเป็นการทดสอบจาก **ServerA** โดยใช้คำสั่ง `ping 8.8.8.8` (ซึ่งเป็น Public DNS ของ Google) เพื่อตรวจสอบว่าเครื่องเซิร์ฟเวอร์ในเครือข่ายภายในสามารถส่งข้อมูลออกไปยังเครือข่ายภายนอกได้หรือไม่

- **ผลการทดสอบ:** พบว่ามีการตอบกลับ (Reply) ครบทั้ง 4 แพ็กเกจ โดยไม่มีการสูญเสียข้อมูล (**0% loss**) และมีค่าความหน่วงเฉลี่ย (Average latency) อยู่ที่ **11ms**
- **สรุป:** ServerA สามารถเชื่อมต่อกับอินเทอร์เน็ตได้สำเร็จ

2. การตรวจสอบการทำงานของ NAT Overload (PAT) บน Router R1

หลังจากยืนยันว่าออกอินเทอร์เน็ตได้แล้ว ขั้นตอนต่อมาคือการตรวจสอบที่ **Router R1** ซึ่งทำหน้าที่เป็นทางผ่าน โดยใช้คำสั่ง `show ip nat translations` เพื่อดูตารางการแปลงที่อยู่

- **วัตถุประสงค์:** เพื่อยืนยันว่า Router กำลังใช้เทคนิค **NAT Overload** (หรือ PAT - Port Address Translation) ในการแปลง Private IP ของเครื่องภายใน ให้เป็น Public IP เพื่อออกสู่อินเทอร์เน็ต
- **การวิเคราะห์:** เมื่อมีการ Ping เกิดขึ้น Router จะสร้าง Mapping ในตาราง NAT เพื่อจับคู่ IP ภายในกับ IP ภายนอก พร้อมระบุหมายเลขพอร์ตหรือ Identifier เพื่อให้ข้อมูลขากลับส่งไปถึงเครื่องต้นทางได้อย่างถูกต้อง

🔎 Verification Commands

On R1

None

```
show ip route
show ip nat translations
show ip interface brief
```

show ip route

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       100.10.10.0/30 is directly connected, Serial0/3/0
L       100.10.10.1/32 is directly connected, Serial0/3/0
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/1
L       192.168.10.1/32 is directly connected, GigabitEthernet0/1
S    192.168.20.0/24 [1/0] via 100.10.10.2
     200.1.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.1.1.0/24 is directly connected, GigabitEthernet0/0
L       200.1.1.1/32 is directly connected, GigabitEthernet0/0
S*   0.0.0.0/0 is directly connected, GigabitEthernet0/0
                [1/0] via 200.1.1.2
```

show ip nat translations

```
R1#show ip nat translations
R1#
```

show ip interface brief

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       100.10.10.0/30 is directly connected, Serial0/3/0
L       100.10.10.1/32 is directly connected, Serial0/3/0
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/1
L       192.168.10.1/32 is directly connected, GigabitEthernet0/1
S    192.168.20.0/24 [1/0] via 100.10.10.2
     200.1.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.1.1.0/24 is directly connected, GigabitEthernet0/0
L       200.1.1.1/32 is directly connected, GigabitEthernet0/0
S*   0.0.0.0/0 is directly connected, GigabitEthernet0/0
               [1/0] via 200.1.1.2

R1#show ip nat translations
R1#
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     200.1.1.1       YES manual up                    up
GigabitEthernet0/1     192.168.10.1    YES manual up                    up
GigabitEthernet0/2     unassigned      YES NVRAM  administratively down down
Serial0/0/0            unassigned      YES unset  down                  down
Serial0/0/1            unassigned      YES unset  down                  down
Serial0/1/0            unassigned      YES unset  down                  down
Serial0/1/1            unassigned      YES unset  down                  down
Serial0/2/0            unassigned      YES unset  down                  down
Serial0/2/1            unassigned      YES unset  down                  down
Serial0/3/0            100.10.10.1     YES manual up                    up
Serial0/3/1            unassigned      YES unset  down                  down
Vlan1                  unassigned      YES NVRAM  administratively down down
R1#
```

On R2

None

```
show ip route
ping 100.10.10.1
```

show ip route

```
R2>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 100.10.10.1 to network 0.0.0.0

     100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       100.10.10.0/30 is directly connected, Serial0/0/0
L       100.10.10.2/32 is directly connected, Serial0/0/0
S    192.168.10.0/24 [1/0] via 100.10.10.1
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/1
L       192.168.20.1/32 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 100.10.10.1
```

ping 100.10.10.1

```
R2>ping 100.10.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.10.10.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 6/10/21 ms

R2>
```